Aarón Fernández Ramón

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Education

University of the Balearic Islands — September 2021 - July 2025
Bachelor's Degree in Computer Engineering — *Palma de Mallorca, Balearic Islands*

- **Relevant coursework:** Data Structures and Algorithms (Java, C and Go), Probability and Statistics for Computer Science (Python and R), Programming II (Java), Computational Applications of Linear Algebra (Python and MATLAB), Machine Learning (Java), Artificial Intelligence (Python), Databases II (SQL), Intelligent Systems (Java), Operating Systems (C), English B2.

Alfonso X el Sabio University — Expected completion July 2026
Master's Degree in Artificial Intelligence — *Madrid, Spain*

- **Relevant coursework:** Mathematics and Statistics for AI (Python), AI Techniques: Classification, Clustering, Regression and Deep Learning (Python), Programming and Work Environments for AI.

Experience

Gemminis — April 2025-Present
AI Programmer and Artificial Intelligence Technician — *Palma de Mallorca, Balearic Islands*

- Designed and implemented computer vision systems based on semantic segmentation, instance segmentation, object detection and panoptic models for urban environments and infrastructure analysis.
- Built automated inspection solutions, including height detection for road restraint systems and railway collision-risk analysis using virtual clearance gauges.
- Designed custom VLM models for contextual image validation and AI agent ecosystems with persistent memory, contextual retrieval and dynamic decision-making.

Projects

Real-time pedestrian risk detection system for moving vehicles (Bachelor's Thesis) | *Python, PyTorch, NumPy*

- Developed a real-time risk detection system based on three AI models: instance segmentation (YOLOv8-seg), semantic segmentation (DeepLabv3+) and classification (YOLOv8).
- Classified individuals according to characteristics and vulnerability profiles, adapting risk analysis to different pedestrians in urban environments.
- Optimized an asynchronous pipeline for real-time execution, reducing bottlenecks and computational cost.

AI-powered web prototypes and applications | *Personal project, React, Backend, AI*

- Designed and developed web page and application prototypes with interactive interfaces, information management systems and AI-based functionality.
- Explored the integration of intelligent models, backend services and user experiences accessible through web environments.

AI-based eye-tracking system | *Personal project, Python, Deep Learning, Generative AI*

- Designed and implemented a personal eye-tracking system using deep learning techniques.
- Integrated generative AI models for image enhancement and preprocessing/postprocessing techniques to improve accuracy, stability and reliability of visual analysis.

Panoptic segmentation system for urban inventories (Master's Thesis) | *Python, PyTorch, Computer Vision*

- Developed a panoptic segmentation system at Alfonso X el Sabio University to identify, segment and classify public-road elements.
- Focused on automating digital urban inventories and detecting defects in infrastructure through advanced visual analysis.

Custom VLM for contextual image validation | *Python, Vision-Language Models, Deep Learning*

- Designed and trained a vision-language model specialized in validating images through visual reasoning and interpretable justification generation.
- Analyzes visual evidence, determines detection validity and produces contextual explanations supporting each decision.

AI ecosystem with persistent memory | *Python, AI Agents, Retrieval*

- Developed an agent ecosystem capable of preserving knowledge across interactions through persistent memory, contextual retrieval and dynamic information structuring.
- Ongoing project with potential to integrate multimodal models, external tools, continual learning and more advanced decision workflows.

Human-TrAIts | *Spring Boot, TensorFlow, PyTorch, React, Axios, phpMyAdmin*

- Developed a web platform for an AI competition, enabling interactive debates on different topics.
- Designed the full system architecture: React frontend, Spring Boot backend and relational database, integrating external APIs to manage interactions.

Recognitions

Bachelor's Thesis 2025

Third prize for Best Computer Engineering Bachelor's Thesis *University of the Balearic Islands*

- Developed a hybrid artificial intelligence system for detecting risks associated with vulnerable pedestrians in urban roads.
- Media coverage through interviews in *Ultima Hora* and on *Mallorca al Dia* by IB3 Radio.

Certificates

Python for Data Science IBM

Online *July 2024*

- Classification, regression and clustering models with scikit-learn; data manipulation in JSON, XLSX and XLS formats.

Fine-tuning Language Models Hugging Face

Online *August 2025*

- Construction of LLM-based models and manipulation of NLP components.

Transformers Architecture Hugging Face

Online *August 2025*

- Model manipulation and development using Transformer-based architectures.

Technical Skills

Languages: Python, R, Java, C/C++, JavaScript, Go, SQL, HTML/CSS, PHP, Lisp, Octave, MATLAB, Easy68K
Frameworks / Libraries: PyTorch, TensorFlow, Spring Boot, Flask, Jinja2, React, NumPy, Pandas, OpenCV
Tools / Platforms: RStudio, Colab, Jupyter, QGIS, Git/GitHub, NetBeans, Visual Studio, Android Studio, IntelliJ IDEA, Blender, Unity, RabbitMQ, Trello, Jira, Postman, LaTeX
Concepts: Machine Learning, Deep Learning, Neural Networks, APIs, Data Analysis, Statistics, Computer Vision, Databases, Advanced Algorithms, Pipelines, LLMs, VLMs, Graphics Computing, Operating Systems, Agile, Scrum, Kanban

Additional Experience

Coaliment: Cashier and stock replenisher, weekends and holidays (July 2023-Present).
BipBip: Cashier and warehouse duties (June 2022-August 2022).
McDonald's: Kitchen assistant and general support worker (June 2021-September 2021).

Languages

- Spanish: native
- Catalan: B2
- English: B2
- German: basic

Other

- Driving licence: B